MUST PRACTICE

BASIC JAVA PROGRAMS FOR BEGINNERS

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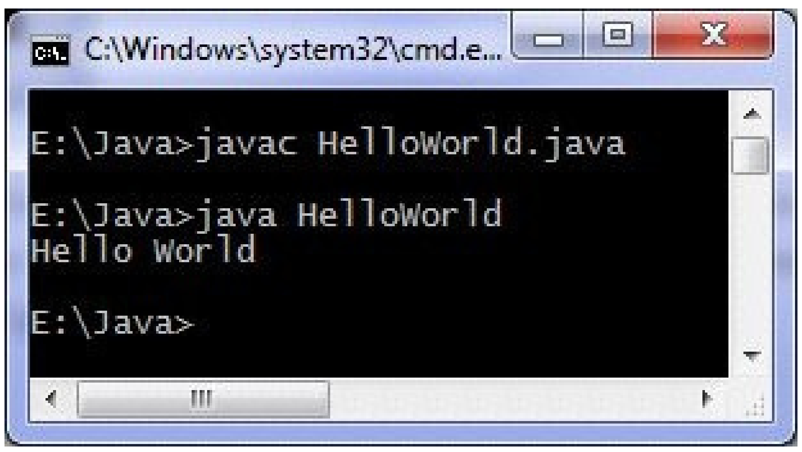
1. Hello World example

```
class HelloWorld {
public static void main(String args[]) {

System.out.println("Hello World"); }
}
```

"Hello World" is passed as an argument to println method, you can print whatever you want. There is also a print method which doesn't takes the cursor to beginning of next line as println does. System is a class, out is object of PrintStream class and println is the method.

Output of program:

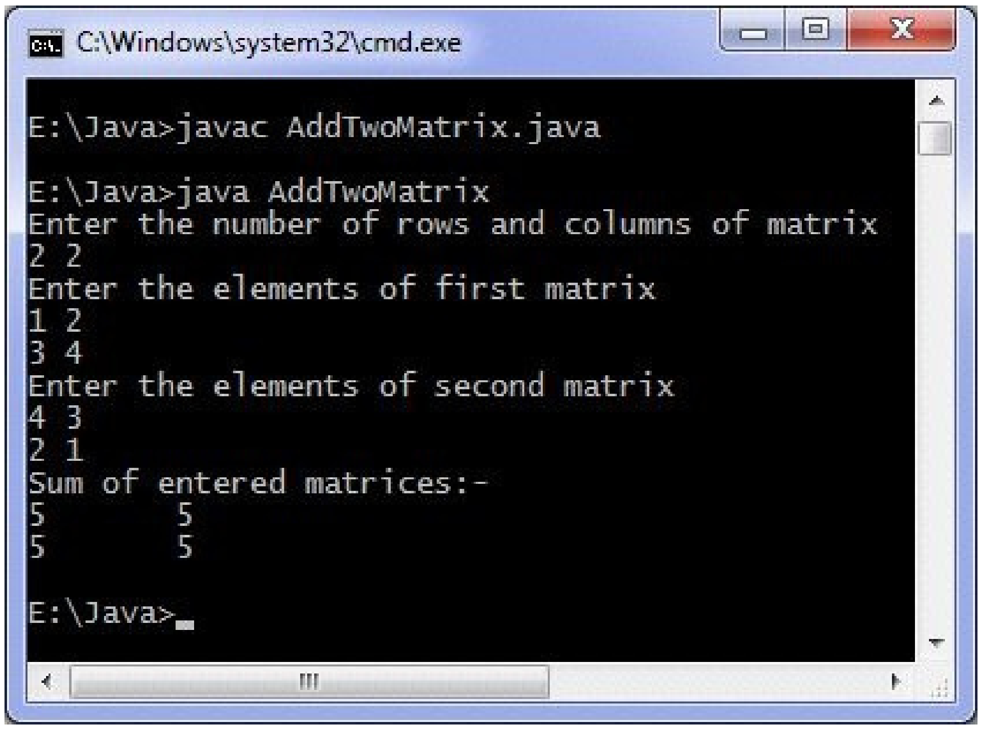


2. Add Two Matrices

```
import java.util.Scanner;
class AddTwoMatrix {
public static void main(String args[]) {
Scanner in = new Scanner(System.in);

System .out.println("Enter
the number of rows and
columns of matrix");
```

```
m = in.nextInt();
n = in.nextInt();
int first[][] = new int[m][n];
int second[][] = new int[m][n]; int sum[][] = new int[m][n];
System .out.println("Enter
the elements of first
matrix");
for( c=0;c<m; c++ )
for ( d = 0 ; d < n ; d++ )
int m, n, c, d;
first[c][d] = in.nextInt();
System .out.println("Enter the elements of second matrix");
for ( c = 0 ; c < m ; c++ )
for ( d = 0 ; d < n ; d++ )
second[c][d] = in.nextInt(); for ( c = 0 ; c < m ; c++ )
for ( d = 0 ; d < n ; d++ )
sum [c][d] = first[c][d] + second[c]
[d]; //replace '+' with '-' to subtract matrices
System .out.println("Sum of entered matrices:-");
for ( c = 0 ; c < m ; c++ )
{
for ( d = 0 ; d < n ; d++ )
System.out.print(sum[c][d] +"\t");
System .out.println();
}
}
}
```

Output of program:



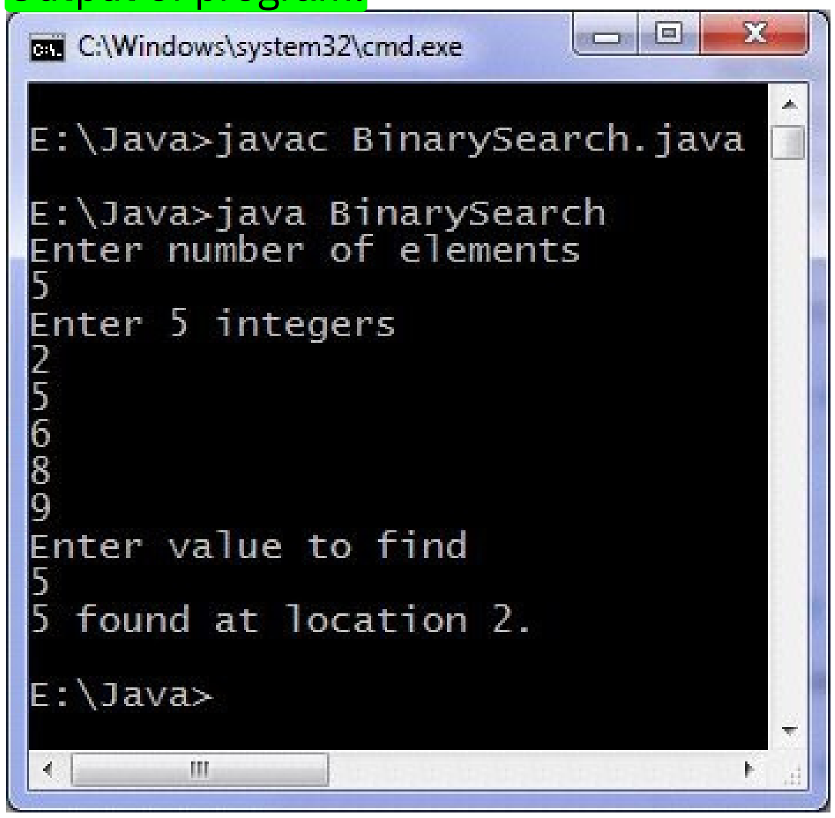
This code adds two matrix, you can modify it to add any number of matrices. You can create a Matrix class and create it's objects and then create an add method which sum the objects, then you can add any number of matrices by repeatedly calling the method using a loop.

3. Binary Search

```
import java.util.Scanner;
class BinarySearch {
public static void main(String args[]) {

int c, first, last, middle, n, search, array[];
Scanner in = new Scanner(System.in);
System.out.println("Enter
number of elements");
n = in.nextInt();
array = new int[n];
System.out.println("Enter "
+ n + " integers");
for (c = 0; c < n; c++) array[c] =

in. nextInt();
System.out.println("Enter
value to find");

search = in.nextInt();
first = 0; last =n-1;
middle = (first + last)/
2;

if ( array[middle] < search ) first = middle + 1; else if

( array[middle] == search ) {
System.out.println(search +
```

```
" found at location " +
(middle + 1) + ".");
break; }
else
last = middle - 1;
middle = (first +
last)/2;
}
if ( first > last )
System.out.println(search + " is not present in the
list.\n");
}
}
while( first <= last ) {
```

Output of program:



Other methods of searching are Linear search and Hashing. There is a binarySearch method in Arrays class which can also be used.

```
import java.util.Arrays;
class BS {
public static void main(String args[])
{
char characters[] =
{ 'a', 'b', 'c', 'd', 'e' };
System.out.println(Arrays.bi
narySearch(characters,
'a'));
```

```
System.out.println(Arrays.bi
narySearch(characters,
'p'));
}

}
```

binarySearch method returns the location if a match occurs otherwise - (x+1) where x is the no. of elements in the array, For example in the second case above when p is not present in characters array the returned value will be -6.

4. Armstrong number

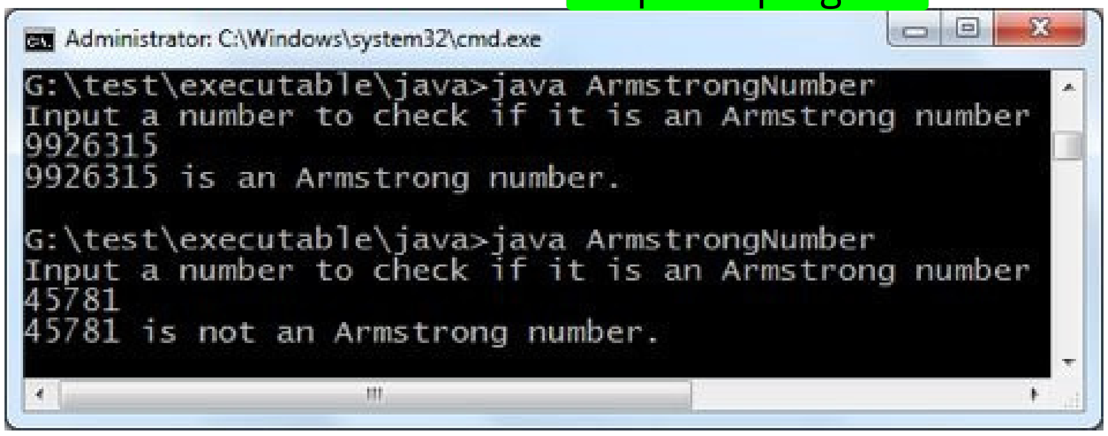
This java program checks if a number is Armstrong or not. Armstrong number is a number which is equal to sum of digits raise to the power total number of digits in the number. Some Armstrong numbers are: 0, 1, 4, 5, 9, 153, 371, 407, 8208 etc.

Java programming code

```
import java.util.Scanner;
class ArmstrongNumber {
public static void main(String args[]) {

int n, sum = 0, temp, remainder, digits = 0;
Scanner in = new Scanner(System.in);

System .out.println("Input a number to check if it is an
Armstrong number"); n = in.nextInt(); temp = n;
// Count number of digits

while (temp != 0) {
%10;
digits++;

temp = temp/10;
}
temp = n;
while (temp != 0) {

remainder = temp
sum = sum + power(remainder, digits);
temp = temp/10;
}
if (n == sum)

System .out.println(n + " is an Armstrong number.");
else

System .out.println(n + " is not an Armstrong number.");
}

static int power(int n, int r) { int c, p = 1;
```

```
for (c = 1; c <= r; c+
+)
return p;
p = p*n; }
}
```

[redacted] Output of program:



Using one more loop in the above code you can generate Armstrong numbers from 1 to n(say) or between two integers (a to b).

5.Bubble sort

Java program to bubble sort: This code sorts numbers inputted by user using Bubble sort algorithm.

Java programming code

```
import java.util.Scanner;
class BubbleSort { public static void

main(String []args) { int n, c, d, swap; Scanner in = new

Scanner (System.in);
System.out.println("Input
number of integers to
sort");
n = in.nextInt(); int array[] = new

int[n]; System.out.println("Enter " + n + " integers");
for (c = 0; c < n; c++) array[c] = in.nextInt();
for (c = 0; c < ( n - 1 ); c++) {

for (d = 0; d < n - c
- 1; d++) {
if (array[d] >

array[d+1]) /* For descending order use < */

{
```

```
array[d];
swap =

array [d] =
array[d+1];
array[d+1] = swap;
}
} }

System .out.println("Sorted
list of numbers");
for (c = 0; c < n; c++)

System .out.println(array[c])
;
}

}
```

Complexity of bubble sort is O(n2) which makes it a less frequent option for arranging in sorted order when quantity of numbers is high.

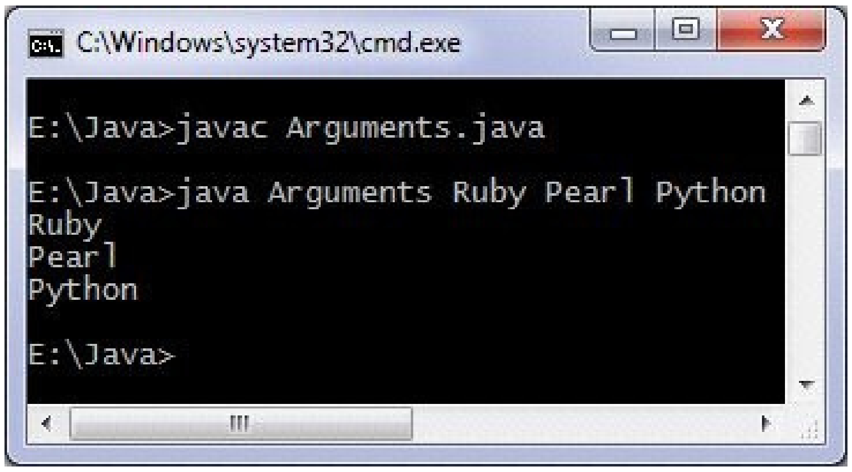
Output of program:

6. Command line arguments

```
class Arguments { public static void main(String[] args) {
for (String t: args) {
System.out.println(t);
}

}
```

```
}
```

Output :



7. Find Odd or Even

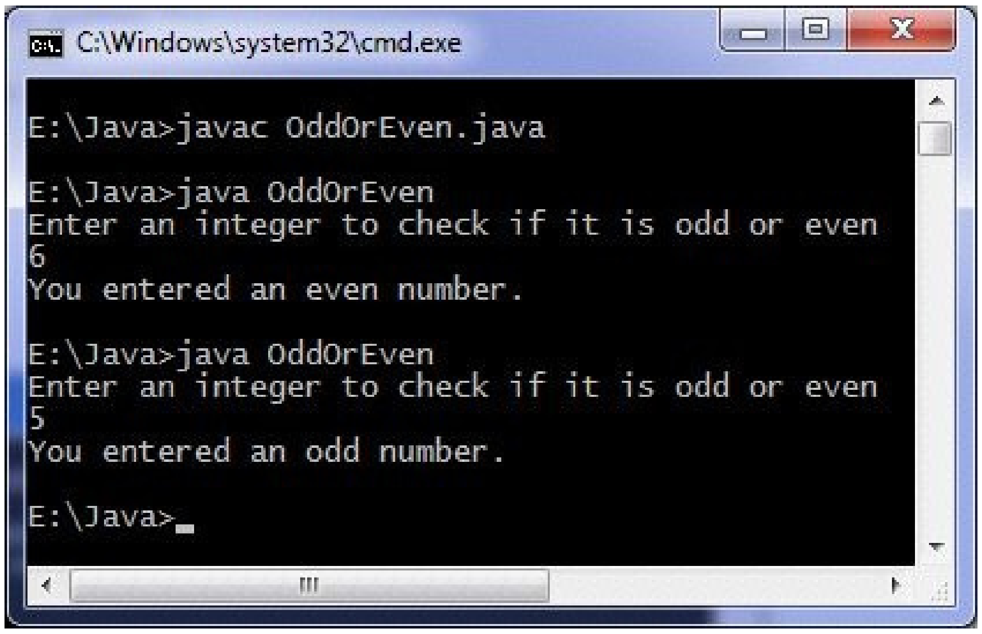
This java program finds if a number is odd or even. If the number is divisible by 2 then it will be even, otherwise it is odd. We use modulus operator to find remainder in our program.

Java programming source code

```
import java.util.Scanner;
class OddOrEven {
public static void main(String args[]) {

System .out.println(“Enter an
integer to check if it is
odd or even ”);
Scanner in = new Scanner(System.in);

x = in.nextInt();
if ( x % 2 == 0 )

System .out.println(“You
entered an even number.”);
else

System .out.println(“You
entered an odd number.”);
}

int x;
}
```

[redacted]

Output of program:



Another method to check odd or even, for explanation see:

```
import java.util.Scanner;
class EvenOdd {
public static void main(String args[]) {
int c;

System .out.println("Input an
integer");
Scanner in = new Scanner(System.in);

c = in.nextInt(); if ( (c/2)*2 == c )
System.out.println("Even"); else

System .out.println("Odd");
}
}
```

There are other methods for checking odd/even one such method is using bitwise operator.

8. convert Fahrenheit to Celsius

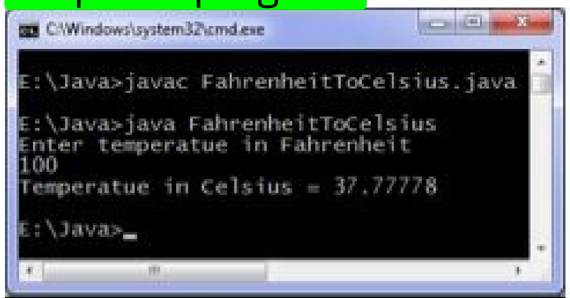
Java program to convert Fahrenheit to Celsius: This code does temperature conversion from Fahrenheit scale to Celsius scale.

Java programming code

```
import java.util.*;

class FahrenheitToCelsius { public static void
main(String[] args) { float temperatue; Scanner in = new

Scanner (System.in);
System.out.println("Enter
temperatue in Fahrenheit");

temperatue =
in.nextInt();
temperatue =
```

```
((temperatue - 32)*5)/9;
System.out.println("Temperat
ue in Celsius = " +
temperatue);
}

}
```


Output of program:



[redacted]

[redacted]

For Celsius to Fahrenheit conversion
use
T = 9*T/5 + 32

where T is temperature on Celsius scale. Create and test Fahrenheit to Celsius program yourself for practice.

9. Display Date and Time

Java program to display date and time, print date and time using java program

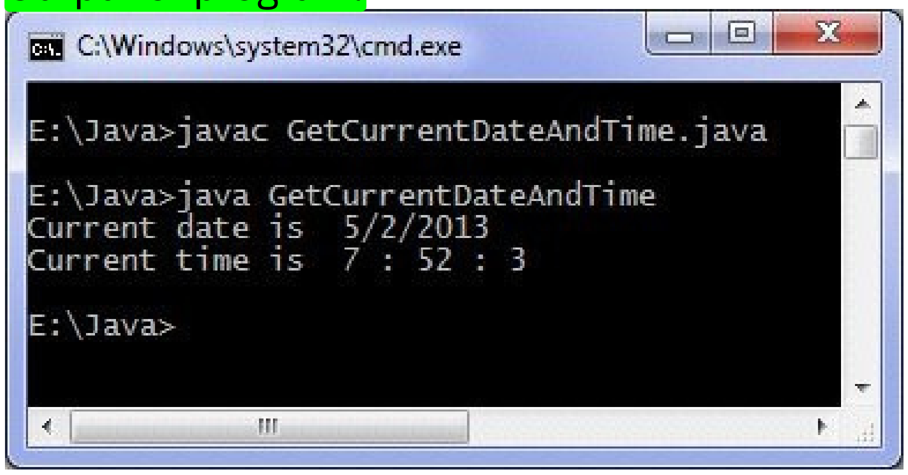
Java date and time program :- Java code to print or display current system date and time. This program prints current date and time. We are using GregorianCalendar class in our program. Java code to print date and

time is given below :-

```
import java.util.*;
class GetCurrentDateAndTime {
public static void main(String args[]) {
int second, minute, hour;

GregorianCalendar date = new
GregorianCalendar();

day =
date.get(Calendar.DAY_OF_MON
TH);

int day, month, year;

month =
date.get(Calendar.MONTH);
```

```
year =
date.get(Calendar.YEAR);
second =
date.get(Calendar.SECOND);
minute =
date.get(Calendar.MINUTE);
hour =
date.get(Calendar.HOUR); System.out.println("Current date is
"+day+"/"+(month +1)+"/"+year);
System.out.println("Current time is "+hour+" : "+minute +" :
"+second);
}

}
```

Output of program:



Don't use Date and Time class of java.util package as their methods are deprecated means they may not be supported in future versions of JDK. As an alternative of GregorianCalendar class you can use Calendar class.

10. Largest of three integers

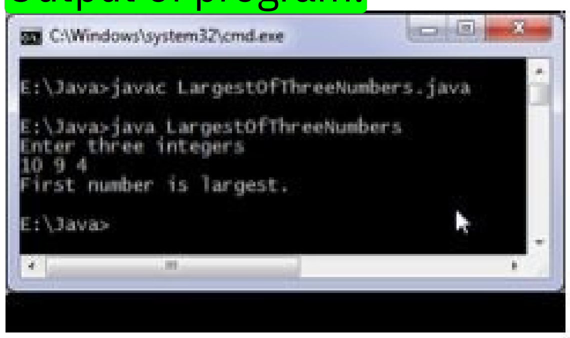
This java program finds largest of three numbers and then prints it. If the entered numbers are unequal then "numbers are not distinct" is printed.

Java programming source code

```
import java.util.Scanner;
class LargestOfThreeNumbers {
public static void main(String args[]) {

System .out.println("Enter
three integers ");
Scanner in = new Scanner(System.in);

x = in.nextInt();
y = in.nextInt();
z = in.nextInt();
```

```
if ( x > y && x > z )
System .out.println(“First
number is largest.”);
else if ( y > x && y > z)
System .out.println(“Second
number is largest.”);
int x, y, z;
else if ( z > x && z > y)
System .out.println(“Third number is largest.”);
else
System .out.println(“Entered numbers are not distinct.”); }
}
```

Output of program:



11. Java - Sample Programs

Program 1
Find Maximum of 2 nos.

```
class Maxof2{
public static void main(String args[]){ //taking value as command line
argument.
//Converting String format to
Integer value
int i = Integer.parseInt(args[0]); int j = Integer.parseInt(args[1]); if(i > j)
System.out.println(i+” is greater than “+j);
else
System.out.println(j+” is greater
than “+i); }
}
```

Program 2

Find Minimum of 2 nos. using conditional operator

```
class Minof2{
public static void main(String args[]){
```

```
//taking value as command line argument.
//Converting String format to Integer value
int i = Integer.parseInt(args[0]); int j = Integer.parseInt(args[1]); int result = (i<j)?i:j; System.out.println(result+" is a
minimum value"); }
}
```

Program 3

Write a program that will read a float type value from the keyboard and print the following output.

->Small Integer not less than the number.
->Given Number.
->Largest Integer not greater than the number.

```
class ValueFormat
{
public static void main(String args[]){ double i = 34.32; //given number System.out.println("Small Integer
not greater than the number : "+Math.ceil(i));
System.out.println("Given Number : "+i);
System.out.println("Largest Integer not greater than the number : "+Math.floor(i));
```

/*Write a program to generate 5 Random nos. between 1 to 100, and it should not follow with decimal point.
***/**

```
class RandomDemo{ public static void main(String args[]){
for(int i=1;i<=5;i++){ System.out.println((int)
(Math.random()*100)); }
} }
```

Program 5

/* Write a program to display a greet message according to Marks obtained by student. */

```
class SwitchDemo{
public static void main(String
args[]){
int marks =
Integer.parseInt(args[0]); // take marks as command line argument.
switch(marks/10){ case 10:
case 9:
case 8: System.out.println("Excellent"); break;
case 7: System.out.println("Very
Good"); break;
case 6:
System.out.println("Good"); break;
case 5: System.out.println("Work
Hard");
break;
case 4:
System.out.println("Poor"); break;
case 3: case 2:
case 1: case 0:
System.out.println("Very
```

```
Poor");
break;
default:
System.out.println("Invalid value Entered"); } }
```

/*Write a program to find SUM AND PRODUCT of a given Digit. */

```
class Sum_Product_ofDigit{
public static void main(String args[]){
int num = Integer.parseInt(args[0]); //taking value as command line argument.
int temp = num,result=0; //Logic for sum of digit while(temp>0){ result = result + temp;
temp–; }
System.out.println("Sum of Digit for "+num+" is : "+result);
//Logic for product of digit temp = num;
result = 1;
while(temp > 0){
result = result * temp;
temp–; }
System.out.println("Product of

Digit for "+num+" is : "+result); }
```

/*Write a program to Find Factorial of Given no. */ class Factorial{

```
public static void main(String args[]){
int num = Integer.parseInt(args[0]); // take argument as command line
int result = 1; while(num>0){
result = result * num;
num–; }
System.out.println("Factorial of Given no. is : "+result); } }
```

Program 8

/*Write a program to Reverse a given no. */

```
class Reverse{
public static void main(String args[]){
int num = Integer.parseInt(args[0]); // take argument as command line
int remainder, result=0; while(num>0){
remainder = num%10;
result = result * 10 + remainder;
num = num/10;

}
System.out.println("Reverse number is : "+result);
} }
```

Program 9

/*Write a program to find Fibonacci series of a given no.

Example : Input - 8

Output - 1 1 2 3 5 8 13 21 */

```
class Fibonacci{
public static void main(String
args[]){
int num =
Integer.parseInt(args[0]); //taking no. as command line argument. System.out.println("*****Fibonac
ci Series*****");
int f1, f2=0, f3=1;
for(int i=1;i<=num;i++){ System.out.print(" "+f3+" "); f1 = f2;  f2 = f3;
```

```
f3 = f1 + f2;
} }
}
```

Program 10

/* Write a program to find sum of all integers greater than 100 and less than 200 that are divisible by 7 */

```
class SumOfDigit{

public static void main(String args[]){
int result=0;
for(int i=100;i<=200;i++){
if(i%7==0) result+=i;
}
System.out.println("Output of Program is : "+result); } }
```

Program 11

/* Write a program to Concatenate string using for Loop Example: Input - 5 Output - 1 2 3 4 5 */

```
class Join{
public static void main(String
args[]){
int num = Integer.parseInt(args[0]); String result = " "; for(int i=1;i<=num;i++){
result = result + i + " "; }
System.out.println(result); }
}
```

Program 12

/* Program to Display Multiplication Table */

```
class MultiplicationTable{
public static void main(String args[]){

int num = Integer.parseInt(args[0]);
System.out.println("*****MULTIPLI CATION TABLE*****"); for(int i=1;i<=num;i++){ for(int j=1;j<=num;j++){
System.out.print(" "+i*j+" "); } System.out.print("\n"); } } }
```

Program 13

/* Write a program to Swap the values */

```
class Swap{
public static void main(String args[]){
int num1 = Integer.parseInt(args[0]);
int num2 = Integer.parseInt(args[1]);
System.out.println("\n***Before Swapping***"); System.out.println("Number 1 : "+num1);
System.out.println("Number 2 : "+num2);
//Swap logic
num1 = num1 + num2; num2 = num1 - num2;
num1 = num1 - num2; System.out.println("\n***After
Swapping***"); System.out.println("Number 1 : "+num1); System.out.println("Number 2 :
"+num2);

} }
```

Program 14

/* Write a program to convert given no. of days into months and days.

(Assume that each month is of 30 days)
Example : Input - 69
Output - 69 days = 2 Month and 9 days */

```
class DayMonthDemo{
public static void main(String
args[]){
int num = Integer.parseInt(args[0]); int days = num%30; int month = num/30; System.out.println(num+" days =
"+month+" Month and "+days+" days");
} }
```

Program 15

/*Write a program to generate a Triangle.

eg:

1
22
333
4 4 4 4 and so on as per user given

```
number */ class Triangle{
public static void main(String args[]){
int num = Integer.parseInt(args[0]);

for(int i=1;i<=num;i++){ for(int j=1;j<=i;j++){
System.out.print(" "+i+" "); }
System.out.print("\n"); }
} }
```

Program 16

/* Write a program to Display Invert Triangle.

Example: Input - 5

Output : 55555 4444 333 22
1 */

```
class InvertTriangle{
public static void main(String
args[]){
int num =
Integer.parseInt(args[0]); while(num > 0){
for(int j=1;j<=num;j++){ System.out.print(" "+num+"
");
}
System.out.print("\n");
num—; }
```

/*Write a program to find whether given no. is Armstrong or not.
Example :
Input - 153
Output - 1^3 + 5^3 + 3^3 = 153, so it is Armstrong no. */ class Armstrong{

```
public static void main(String args[]){
int num = Integer.parseInt(args[0]);
int n = num; //use to check at last time
int check=0,remainder; while(num > 0){
remainder = num % 10;
check = check + (int)Math.pow(remainder,3);
num = num / 10; }
if(check == n) System.out.println(n+" is an
Armstrong Number"); else
System.out.println(n+" is not a Armstrong Number");
```

/* Write a program to Find whether number is Prime or Not. */ class PrimeNo{

```
public static void main(String args[]){
int num = Integer.parseInt(args[0]);
int flag=0;
for(int i=2;i<num;i++){

if(num%i==0) {
System.out.println(num+" is not a Prime Number");
flag = 1;
break; }
} if(flag==0)
System.out.println(num+" is a Prime Number");
} }
```

Program 19

/* Write a program to find whether no. is palindrome or not. Example :
Input - 12521 is a palindrome
no.
Input - 12345 is not a
palindrome no. */

```
class Palindrome{
public static void main(String
args[]){
int num =
Integer.parseInt(args[0]);
int n = num; //used at last time
check
int reverse=0,remainder;
while(num > 0){
remainder = num % 10; reverse = reverse * 10 +
remainder;
num = num / 10;
}
if(reverse == n)
System.out.println(n+" is a

Palindrome Number"); else
System.out.println(n+" is not a Palindrome Number"); } }
```

Core Java Programs [PAGE 3]

Some Java programs which help lot of java beginners to understand the basic fundamentals in Java programming. Most of these programs take input from the command line. *Ex - int num = Integer.parseInt(args[0]);*

/* switch case demo

Example : Input - 124
Output - One Two Four */

```
class SwitchCaseDemo{
public static void main(String
args[]){ try{
int num = Integer.parseInt(args[0]); int n = num; //used at last time check int reverse=0,remainder; while(num > 0){
remainder = num % 10;
reverse = reverse * 10 + remainder; num = num / 10; }
String result=""; //contains the actual output while(reverse > 0){
```

```
“Zero “;
“One “;
“Two “;
“Three “;
“Four “;
“Five “;
“;
“Seven “;
“Eight “;
remainder = reverse % 10; reverse = reverse / 10; switch(remainder){
case 0 :
result = result +
break; case 1 :
result = result +
break; case 2 :
result = result +
break; case 3 :
result = result +
break; case 4 :
result = result +
break; case 5 :
result = result +
break; case 6 :
result = result + “Six
break; case 7 :
result = result +
break; case 8 :
result = result +
break; case 9 :
result = result +
break; default:
result=””; }
}
System.out.println(result);
}catch(Exception e){ System.out.println(“Invalid
Number Format”); }
} }
```

Program 21
/* Write a program to generate Harmonic Series. Example : Input - 5
Output - 1 + 1/2 + 1/3 + 1/4 + 1/5 = 2.28 (Approximately) */ class HarmonicSeries{

```
public static void main(String args[]){
int num = Integer.parseInt(args[0]); double result = 0.0; while(num > 0){
result = result + (double) 1 / num;
num—; }
System.out.println(“Output of Harmonic Series is “+result); “Nine “;
```

Program 22
/*Write a program to find average of consecutive N Odd no. and Even no. */

```
class EvenOdd_Avg{
public static void main(String args[]){
int n = Integer.parseInt(args[0]);
int cntEven=0,cntOdd=0,sumEven=0,sum Odd=0;
while(n > 0){ if(n%2==0){
cntEven++;
sumEven = sumEven + n; }
else{
```

```
cntOdd++;
sumOdd = sumOdd + n;
}
n—; }
int evenAvg,oddAvg;
evenAvg = sumEven/cntEven; oddAvg = sumOdd/cntOdd; System.out.println("Average of first
N Even no is "+evenAvg); System.out.println("Average of first N Odd no is "+oddAvg);
} }
```

Program 23

/* Display Triangle as follow : BREAK DEMO.

1 23
456
7 8 9 10 ... N */ class Output1{

```
public static void main(String
args[]){
int c=0;
int n = Integer.parseInt(args[0]); loop1: for(int i=1;i<=n;i++){ loop2: for(int j=1;j<=i;j++){
if(c!=n){ c++;
System.out.print(c+"
");
}
else
break loop1;
}
System.out.print("\n"); }
```

Program 24

/* Display Triangle as follow

0

10
101
0 1 0 1 */

```
class Output2{
public static void main(String
args[]){
for(int i=1;i<=4;i++){
for(int j=1;j<=i;j++){ System.out.print(((i +j)%2)+" ");
}
System.out.print("\n");
} }
}
```

Program 25

/* Display Triangle as follow

1

24
369
4 8 12 16 ... N (indicates no. of

Rows) */

```
class Output3{
public static void main(String
```

```
args[]){
int n = Integer.parseInt(args[0]);
for(int i=1;i<=n;i++){ for(int j=1;j<=i;j++){
“);
}
} }
System.out.print((i*j)+” System.out.print(“\n”); }
```

12.Java Programs part 2

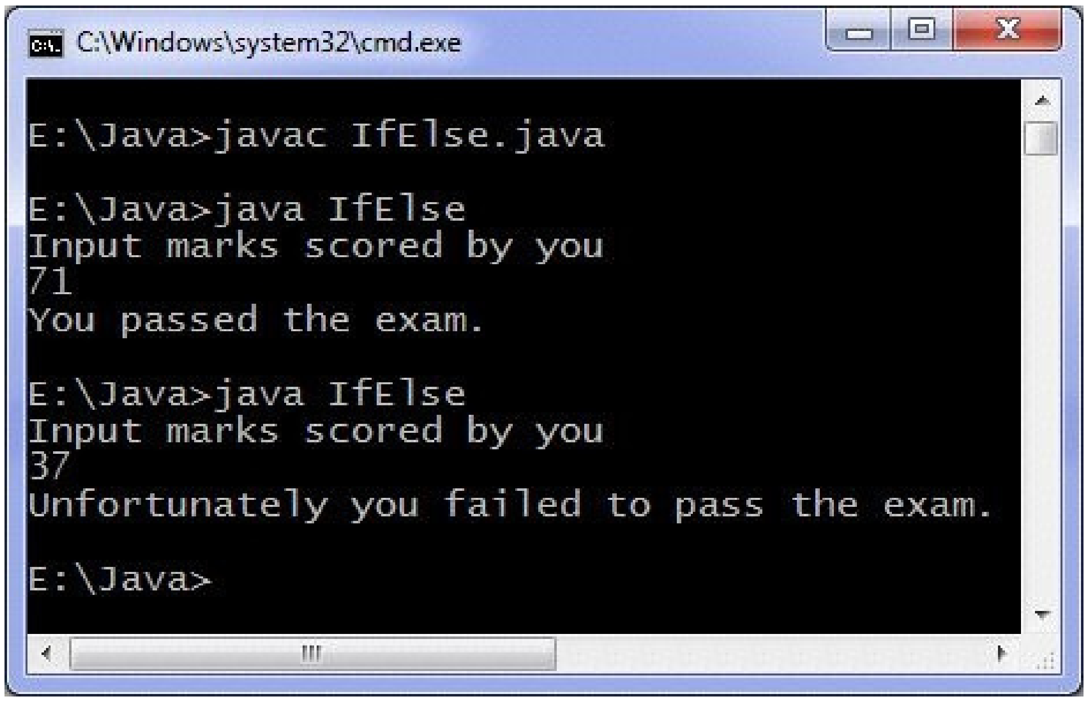
1. Java if else program

Java if else program uses if else to execute statement(s) when a condition is fulfilled. Below is a simple program which explains the usage of if else in java programming language.

Java programming if else
statement

```
// If else in Java code
import java.util.Scanner;

class IfElse { public static void main(String[] args) { int
marksObtained,

passingMarks ;
passingMarks = 40;
Scanner input = new Scanner(System.in);

System.out.println("Input
marks scored by you");
marksObtained =

input. nextInt();
if (marksObtained >=
passingMarks) {
System.out.println("You
passed the exam.");
}
else {

System .out.println("Unfortun
ately you failed to pass the
exam.");
} }

}
```

Output of program:



Above program ask the user to enter marks obtained in exam and the input marks are compared against minimum passing marks. Appropriate message is printed on screen based on whether user passed the exam or not. In the above code both if and else block contain only one statement but we can execute as many statements as

required.

2. Nested If Else statements

You can use nested if else which means that you can use if else statements in any if or else block.

```
import java.util.Scanner;
class NestedIfElse { public static void main(String[] args)
{ int marksObtained, passingMarks; char grade;
passingMarks = 40; Scanner input = new

Scanner (System.in);
System.out.println("Input
marks scored by you");

marksObtained =
input.nextInt();
if (marksObtained >= passingMarks) { if (marksObtained > 90)

grade = 'A';
else if (marksObtained > 75)
else if (marksObtained > 60)
else
grade = 'B';

grade = 'C';
grade = 'D';
System.out.println("You passed the exam and your grade is "
+ grade);
}
else {

grade = 'F';

System .out.println("You failed and your grade is " +
grade);
} }

}
```

Output of program:

C:\Windows\system32\cmd.exe

```
E:\Java>javac NestedIfElse.java

E:\Java>java NestedIfElse
Input marks scored by you
63
You passed the exam and your grade is C

E:\Java>java NestedIfElse
Input marks scored by you
39
You failed and your grade is F

E:\Java>
```

3. Java for loop

Simple for loop example in Java

Example program below uses for loop to print first 10 natural numbers i.e. from 1 to 10.

```
//Java for loop program
class ForLoop { public static void main(String[] args) { int
c;
for (c = 1; c <= 10; c+ +) {
System.out.println(c);

}
}
```

Output of program:

4. Java for loop example to print stars in console

Following star pattern is printed

```
*
 **
 ***
  ****
   *****
```

```
class Stars {
public static void
main(String[] args) {
int row, numberOfStars;
for (row = 1; row <= 10; row++) { for(numberOfStars = 1;
numberOfStars <= row;

numberOfStars ++) {
System.out.print("*");
}

System.out.println(); // Go to next line } }
```

Above program uses nested for loops (for loop inside a for loop) to print stars. You can also use spaces to create another pattern, It is left for

you as an exercise.

Output of program:

5. Java while loop

Java while loop is used to execute statement(s) until a condition holds true. In this tutorial we will learn looping using Java while loop examples. First of all lets discuss while loop syntax:

```
while (condition(s)) {
// Body of loop
}
```

1. If the condition holds true then the body of loop is executed, after execution of loop body condition is tested again and if the condition is

true then body of loop is executed again and the process repeats until condition becomes

false. Condition is always evaluated to true or false and if it is a constant, For example while (c) { …} where c is a constant then any non zero value of c is considered true and zero is considered false.

2. You can test multiple conditions such as

```
while ( a > b && c != 0) { // Loop body }
```

Loop body is executed till value of a is greater than value of b and c is not equal to zero.

3. Body of loop can contain more than one statement. For multiple statements you need to place them in a block using {} and if body of loop contain only single statement you can optionally use {}. It is recommended to use braces always to make your program easily readable and understandable.

Java while loop example

Following program asks the user to input an integer and prints it until user enter 0 (zero).

```
import
java.util.Scanner;
class WhileLoop { public static void main(String[] args) {
int n;
Scanner input = new Scanner(System.in);

System .out.println("Input an
integer");
while ((n = input.nextInt()) != 0) {

System .out.println("You
entered " + n);
System.out.println("Input an
integer");
}

System .out.println("Out of loop");
}

}
```

Output of program:

```
C:\Windows\system32\cmd.exe

E:\Java>javac WhileLoop.java

E:\Java>java WhileLoop
Input an integer
7
You entered 7
Input an integer
-2
You entered -2
Input an integer
9546
You entered 9546
Input an integer
0
Out of loop

E:\Java>_
```

6. Java program to print alphabets

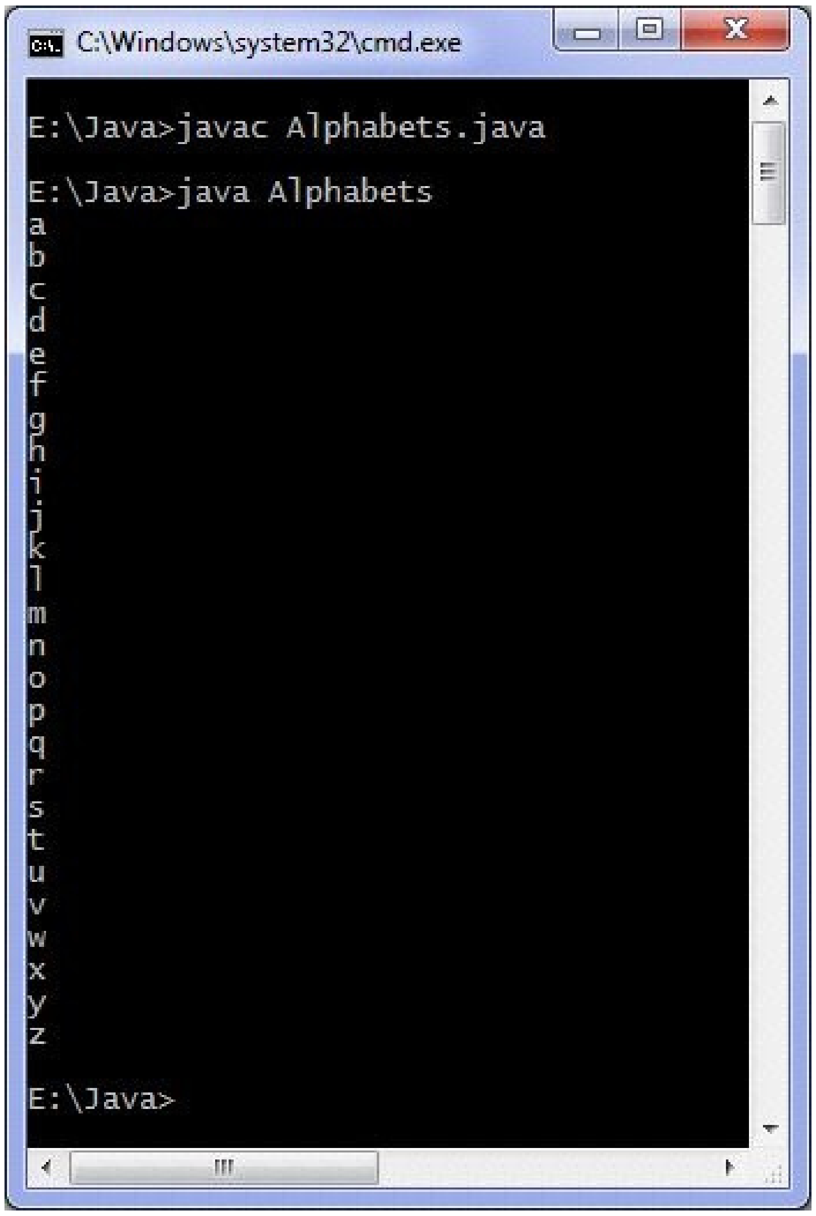
This program print alphabets on screen i.e a, b, c, ..., z. Here we print alphabets in lower case.

Java source code

```
class Alphabets {
public static void main(String args[]) {
'z' ; ch++ )
System.out.println(ch);
char ch;
for( ch = 'a' ; ch <=
}
}
```

You can easily modify the above java program to print alphabets in upper case.

Output of program:



Printing alphabets using while loop

(only body of main method is shown): `char c = ‘a’; while (c <= ‘z’) {`

[redacted]}

```
System.out.println(c);
c++;
```

Using do while loop:

```
char c = ‘A’;
do { System.out.println(c); c++; } while (c <= ‘Z’);
```

Java program to print multiplication table

This java program prints multiplication table of a number

[redacted]

entered by the user using a

for
loop.

[redacted]

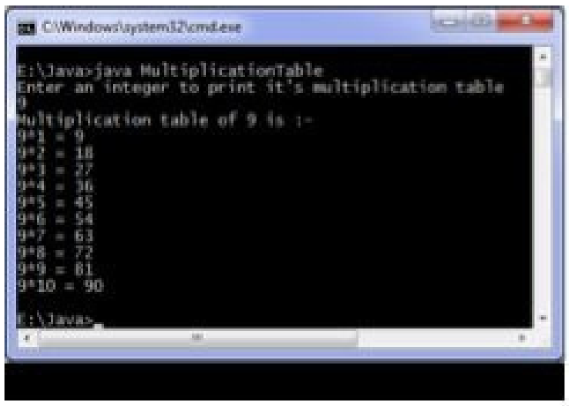
You can modify it for
while
loop for practice.
Java programming source code

```
import java.util.Scanner;
class MultiplicationTable {
public static void main(String args[]) {
int n, c; System.out.println("Enter an
```

while
or
do

```
integer to print it's
multiplication table");
Scanner in = new Scanner(System.in);

n = in.nextInt();
System.out.println("Multipli
cation table of "+n+"
is :-");
for ( c = 1 ; c <= 10 ; c++ )

System.out.println(n+"*"+c+"
= "+(n*c));
}
}
```

Output of program:


C:\Windows\system32\cmd.exe
E:\Java>java MultiplicationTable
Enter an integer to print it's multiplication table
9
Multiplication table of 9 is :-
9*1 = 9
9*2 = 18
9*3 = 27
9*4 = 36
9*5 = 45
9*6 = 54
9*7 = 63
9*8 = 72
9*9 = 81
9*10 = 90
E:\Java>

How to get input from user in java

This program tells you how to get input from user in a java program. We are using Scanner class to get input from user. This program firstly asks the user to enter a string and then the string is printed, then an integer and entered integer is also printed and finall y a float and it is also printed on the screen. Scanner class is present in java.util package so we import this package in our program. We first create an object of Scanner class and then we use the methods of Scanner class. Consider the statement Scanner a = new Scanner(System.in); Here Scanner is the class name, a is the name of object, new keyword is used to allocate the memory and System.in is the input stream. Following methods of Scanner class

are used in the program below :-

1) nextInt to input an integer
2) nextFloat to input a float
3) nextLine to input a string

Java programming source code

```
import java.util.Scanner;
class GetInputFromUser {
public static void main(String args[]) {
Scanner in = new Scanner(System.in);

System.out.println("Enter a
string");
s = in.nextLine();

System .out.println("You
entered string "+s);
System.out.println("Enter an
integer");

a = in.nextInt();
System.out.println("You
entered integer "+a);
System.out.println("Enter a
float");

b = in.nextFloat();
int a; float b; String s;

System .out.println("You
entered float "+b);
}

}
```

Output of program:

There are other classes which can be used for getting input from user and you can also take input from other devices.

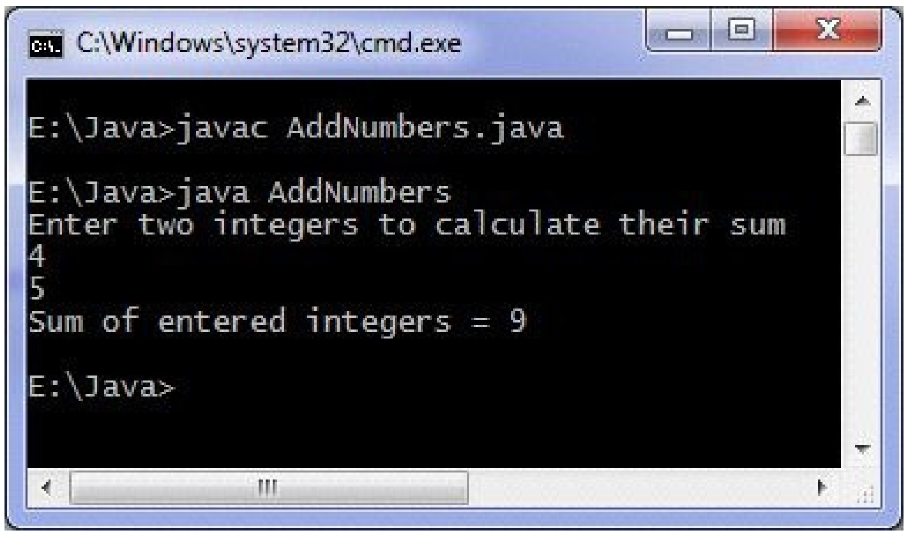
Java program to add two numbers

Java program to add two numbers :-
Given below is the code of java program that adds two numbers which are entered by the user.

Java programming source code

```
import java.util.Scanner;
class AddNumbers {
public static void main(String args[]) {
System .out.println("Enter
two integers to calculate
their sum ");
Scanner in = new Scanner(System.in);
x = in.nextInt();
y = in.nextInt();
z = x + y;
System.out.println("Sum of
entered integers = "+z);
}
}
int x, y, z;
```

Output of program:



Above code can add only numbers in range of integers(4 bytes), if you wish to add very large numbers then you can use BigInteger class. Code to add very large numbers:

```
import java.util.Scanner; import
java.math.BigInteger;
class AddingLargeNumbers { public static void
main(String[] args) {
String number1, number2; Scanner in = new
Scanner(System.in);
System.out.println("Enter
```

```
first large number”);

number1 = in.nextLine();
System.out.println(“Enter
second large number”);

number2 = in.nextLine();
BigInteger first = new
BigInteger(number1);

BigInteger second = new
BigInteger(number2);

BigInteger sum;
sum = first.add(second);
System.out.println(“Result
of addition = ” + sum);
}

}
```

In our code we create two objects of BigInteger class in java.math package. Input should be digit strings otherwise an exception will be raised, also you cannot simply use ‘+’ operator to add objects of BigInteger class, you have to use add method for addition of two objects.
Output of program:

```
Enter first large number 11111111111111
Enter second large number 99999999999999
Result of addition = 111111111111110
```

Java Method example program

```
class Methods {
// Constructor method Methods() {

System .out.println("Construc tor method is called when an
object of it's class is
created");
}

// Main method where
program execution begins
public static void main(String[] args) {

staticMethod();
Methods object = new Methods();

object. nonStaticMethod(); }
// Static method

static void staticMethod() {

System .out.println("Static method can be called without
creating object");
}

// Non static method
void nonStaticMethod() { System.out.println("Non

static method must be called by creating an object"); }

}
```

Output of program:

Java String methods

String class contains methods which are useful for performing operations on String(s). Below program illustrate how to use inbuilt methods of String class.

Java string class program

```
class StringMethods {
public static void main(String args[]) {
int n;

String s = "Java
programming", t = "", u =
"";
System.out.println(s); // Find length of string n =
s.length();

System.out.println("Number
of characters = " + n);

// Replace characters in
string
t = s.replace("Java", "C
++");

System .out.println(s); System.out.println(t); //
Concatenating string

with another string
u = s.concat(" is fun"); System.out.println(s);
System.out.println(u);

}
}
```

Output of program:

13. Java Programs part 3

Java static block program

Java programming language offers a block known as static which is executed before main method executes. Below is the simplest example to understand functioning of static block later we see a practical use of static block.

Java static block program

```
class StaticBlock { public static void

main(String[] args) {
System.out.println(“Main
method is executed.”);
}
static {

System .out.println(“Static
block is executed before
main method.”);
}

}
```

Output of program:

Static block can be used to check conditions before execution of main begin, Suppose we have developed an application which runs only on Windows operating system then we need to check what operating system is installed on user machine. In our java code we check what operating system user is using if user is using operating system other than “Windows” then the program terminates.

```
class StaticBlock { public static void main(String[] args) {

System .out.println(“You
are using Windows_NT
operating system.”);
}

static { String os =
System.getenv(“OS”); if
(os.equals(“Windows_NT”) !=
```

```
true) { System.exit(1);
} }
}
```

We are using getenv method of System class which returns value of environment variable name of which is passed an as argument to it. Windows_NT is a family of operating systems which includes Windows XP, Vista, 7, 8 and others.

Output of program on Windows 7: